

Active-Grade-Six Exclusion for Restart-Four Conjugate Gradients

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September 7, 2026

Abstract

For exact-arithmetic conjugate gradients restarted every four steps, we prove that an initial residual supported on at most six distinct active eigenvalues either terminates or has convergent normalized residual directions along both restart parities. The proof treats the six-node case by a Vandermonde annihilator, which produces a scalar gap parameter, an exact drift identity, and a sign-transport functional. Together with the known eight-dimensional counterexample, this gives

$$7 \leq n_{\min}(4) \leq 8.$$

The original minimum-dimension problem remains unresolved because active grade seven is not treated here.

1 Statement and context

Let A be a real diagonal positive-definite matrix and let r_k be the residual at the beginning of restart block k for exact conjugate gradients with restart length four. Thus k counts completed four-step blocks. When $r_k \neq 0$, write

$$y_k = \frac{r_k}{\|r_k\|_2}.$$

The *active grade* is the number of distinct eigenvalues on whose eigenspaces the initial residual has a nonzero component. Restriction to the corresponding cyclic subspace preserves termination and directional convergence.

Colbrook, Stepaniants, and Townsend proved that for every restart length $s \geq 4$ there is a diagonal positive-definite counterexample of dimension $s + 4$; for $s = 4$ its even normalized residual directions fail to converge [1, Theorem 1.1(ii) and Part C]. They also identified the smallest counterexample dimension as a further question [1, Consequences and further questions]. Define $n_{\min}(4)$ to be that smallest dimension at restart length four.

Our result is the following exact partial answer.

Theorem 1. *For exact-arithmetic restarted conjugate gradients with restart length four on a real diagonal positive-definite matrix, if the initial residual has at most six active distinct eigenvalues, then the iteration either terminates or both normalized residual subsequences*

$$\left(\frac{r_{2j}}{\|r_{2j}\|_2} \right)_{j \geq 0}, \quad \left(\frac{r_{2j+1}}{\|r_{2j+1}\|_2} \right)_{j \geq 0}$$

converge. Consequently,

$$7 \leq n_{\min}(4) \leq 8.$$

The lower bound is the contribution of this note; the upper bound is the cited eight-dimensional construction. No assertion about active grade seven is made.

2 The spectral restart map

After active spectral reduction, let

$$0 < \lambda_1 < \cdots < \lambda_N, \quad w_{k,i} = y_{k,i}^2, \quad \sum_i w_{k,i} = 1.$$

If $N > 4$ and the iteration has not terminated, let P_k be the monic quartic orthogonal to $1, t, t^2, t^3$ for the discrete weights w_k , and put

$$H_k = \sum_i w_{k,i} P_k(\lambda_i)^2 > 0, \quad q_k = P_k P_{k+1}.$$

Because the restart length is even, the signed and squared one-block updates are

$$y_{k+1,i} = \frac{P_k(\lambda_i)}{\sqrt{H_k}} y_{k,i}, \quad w_{k+1,i} = \frac{P_k(\lambda_i)^2}{H_k} w_{k,i}. \quad (1)$$

We use the following degree-independent facts proved in [1, Proposition 1.4 and (G.7)–(G.10)]. Along every nonterminating orbit,

$$H_k \nearrow h > 0, \quad P_{2j} \longrightarrow P_e, \quad P_{2j+1} \longrightarrow P_o, \quad q_k \longrightarrow q_\infty = P_e P_o \quad (2)$$

coefficientwise. Moreover,

$$y_{k+2,i} = \frac{q_k(\lambda_i)}{\sqrt{H_k H_{k+1}}} y_{k,i}, \quad \|y_{k+2} - y_k\|_2 \longrightarrow 0. \quad (3)$$

Every omega-limit state is fixed by the signed two-block map, is supported on between five and eight active nodes, and is supported only at nodes satisfying $q_\infty(\lambda_i) = h$. We use no minimum-dimension conclusion from that source.

3 Six-node algebra

Assume for now that $N = 6$ and that all six weights remain positive. Set

$$\Delta(t) = \prod_{i=1}^6 (t - \lambda_i), \quad D_i = \Delta'(\lambda_i).$$

The four orthogonality equations say that the vector $(w_{k,i} P_k(\lambda_i))_{i=1}^6$ annihilates the first four rows of the Vandermonde matrix. Its nullspace consists of the vectors

$$\left(\frac{c_0 + c_1 \lambda_i}{D_i} \right)_{i=1}^6.$$

The standard Lagrange identities

$$\sum_{i=1}^6 \frac{\lambda_i^j}{D_i} = 0 \quad (0 \leq j \leq 4), \quad \sum_{i=1}^6 \frac{\lambda_i^5}{D_i} = 1$$

and monicity give $\langle P_k, t^4 \rangle_{w_k} = \langle P_k, P_k \rangle_{w_k} = H_k$, hence $c_1 = H_k$. This use of distinct-node evaluation identities is methodologically adjacent to the characteristic-zero relationship between derivative Bézout inversion and multipoint evaluation studied in [2]. That work is logically separate and supplies no restarted-CG dynamical assertion. There is therefore a unique $a_k \in \mathbb{R}$ such that

$$w_{k,i}P_k(\lambda_i) = H_k \frac{\lambda_i - a_k}{D_i}. \quad (4)$$

Combining this with (1) yields

$$w_{k+1,i} = \frac{(\lambda_i - a_k)P_k(\lambda_i)}{D_i}. \quad (5)$$

When the next weights are positive, the values of the monic quintic $(t - a_k)P_k(t)$ alternate in sign at the six nodes. It consequently has one simple zero in each of the five node gaps. The four zeros of P_k occupy four gaps, while a_k occupies the remaining gap.

Applying (4) at times k and $k + 1$ and using (5) shows that the left side below vanishes at every node:

$$(t - a_k)P_kP_{k+1} - H_{k+1}(t - a_{k+1}) = \Delta B_k, \quad (6)$$

where B_k is a monic cubic. Divide

$$B_k = (t - a_k)L_k + u_k,$$

with L_k monic quadratic and $u_k \in \mathbb{R}$, and define the polynomial

$$K_a(t) = \frac{\Delta(t) - \Delta(a)}{t - a}.$$

Evaluation at $t = a_k$, followed by exact polynomial cancellation, gives

$$a_{k+1} - a_k = \frac{\Delta(a_k)}{H_{k+1}} u_k, \quad (7)$$

$$P_kP_{k+1} = H_{k+1} + \Delta L_k + u_k K_{a_k}. \quad (8)$$

Substitution of (8) at the nodes into the two-block energy identity from [1, (G.8)] gives the exact nonnegative increment

$$H_{k+1} - H_k = \frac{u_k^2}{H_{k+1}} \sum_{i=1}^6 w_{k,i} K_{a_k}(\lambda_i)^2. \quad (9)$$

4 Transport of the scalar defect

Suppose first that some omega-limit state has full six-node support. By (2) and the resonance property,

$$P_e P_o = h + \Delta L \quad (10)$$

for a monic quadratic L . At the full-support limit, (4)–(5) show that the same value a parametrizes both phases. Thus a belongs to a node gap $I = (\lambda_g, \lambda_{g+1})$ omitted by the zeros of both P_e and P_o . Coefficient convergence of the factors and the preceding interlacing show that

$$a_k \in I \quad (11)$$

for every sufficiently large k .

For a monic quartic P near either limiting factor, let J_P be the monic quadratic for which the remainder of $J_P \Delta$ modulo P has degree at most one. This quadratic exists uniquely near the limits. Indeed, at a limiting factor, (10) shows that $J_P = L$. A vector in the homogeneous two-by-two system would give $d, e \in \mathcal{P}_1$ such that $P \mid d\Delta - e$. Multiplication by L and reduction using (10) would then give the polynomial identity

$$eL + hd = 0,$$

which forces $d = e = 0$. The system is nonsingular, and J_P depends continuously on P near the two limits.

Define

$$\mathcal{L}_P(f) = [t^3] \text{rem}_P(J_P f).$$

This functional annihilates constants, $P\mathcal{P}_3$, and $\Delta\mathcal{P}_1$. Subtracting (8) at two consecutive times and applying $\mathcal{L}_{P_{k+1}}$ therefore gives

$$u_{k+1}\mathcal{L}_{P_{k+1}}(K_{a_{k+1}}) = u_k\mathcal{L}_{P_{k+1}}(K_{a_k}). \quad (12)$$

At a limiting factor P , let Q denote the opposite factor. The four roots z of P are simple. Lagrange interpolation, together with (10), gives uniformly for $a \in \bar{I}$,

$$\begin{aligned} \mathcal{L}_P(K_a) &= \sum_{P(z)=0} \frac{L(z)}{P'(z)} \frac{\Delta(z) - \Delta(a)}{z - a} \\ &= \frac{h + \Delta(a)L(a)}{P(a)} = Q(a). \end{aligned} \quad (13)$$

Neither limiting factor has a zero on $\bar{I}$. Compactness, continuity, and (13) imply that the two functional values in (12) are nonzero and have the same sign for all large k . Thus either the defects are eventually zero, or $u_{k+1}/u_k > 0$ eventually. In either case u_k has an eventual fixed sign. Since Δ also has a fixed sign on I , the drift identity (7) makes a_k eventually monotone. It is bounded by (11), so a_k converges. Formula (5) and the parity convergence of P_k now imply convergence of both parity weight vectors.

It remains to cover the boundary. If no omega-limit state has full support, every omega-limit state is supported on exactly five nodes. On a fixed five-node support S , with support polynomial Δ_S , the one-dimensional Vandermonde annihilator gives

$$w_i P(\lambda_i) = \frac{h}{\Delta'_S(\lambda_i)} \quad (\lambda_i \in S). \quad (14)$$

For either fixed limiting factor, each of the six possible supports therefore determines at most one limiting weight vector. The omega-limit set of a same-parity sequence is connected because its consecutive step sizes tend to zero by (3); a connected subset of this finite set is a singleton. Hence both parity weight vectors converge in the boundary case as well.

5 Lower grades, signed recovery, and the dimension bound

If a coordinate becomes zero at a finite block, active spectral reduction lowers the grade. Grade at most four terminates. At active grade five, the one-dimensional version of (4) gives

$$w_{k,i} P_k(\lambda_i) = \frac{H_k}{\Delta'_S(\lambda_i)}, \quad w_{k+1,i} = \frac{P_k(\lambda_i)}{\Delta'_S(\lambda_i)}.$$

Applying the same formula at the next block yields $P_k(\lambda_i)P_{k+1}(\lambda_i) = H_{k+1}$ at every active node. Equality in the two-block energy identity gives $H_{k+1} = H_k$, and (3) then gives $y_{k+2} = y_k$. Repeated eigenvalues and initially zero coordinates are already included through active spectral reduction.

For the six-node cases above, we have proved convergence of the squared coordinates along both parities. At every coordinate having positive limiting weight, the omega-limit resonance property and (2) make the signed two-block multiplier in (3) tend to 1. It is therefore positive eventually, so that coordinate's sign stabilizes. Every coordinate with zero limiting weight vanishes in magnitude. Thus the signed vectors y_{2j} and y_{2j+1} converge, proving the first assertion of Theorem 1.

Any instance in dimension at most six has active grade at most six, so it cannot be a counterexample. The diagonal eight-dimensional construction in [1, Part C] supplies the opposite bound, and hence $7 \leq n_{\min}(4) \leq 8$.

6 Limitations and computational disclosure

The argument is an exact-arithmetic exclusion through active grade six. It does not analyze active grade seven, finite-precision restarted CG, or the stability of the asymptotic directions under rounding. The comparison with prior work is deliberately limited to the cited primary paper; no general priority claim is made.

Generative AI assisted proof exploration and drafting. A standard-library Python program using exact rational arithmetic was also used to recompute five sample restart blocks and check the annihilator, polynomial-division, drift, energy, transport, and limiting-functional identities. These finite checks are regression evidence only; the asymptotic proof is the argument given above. The author reviewed the mathematical claims and takes responsibility for the manuscript.

References

- [1] Matthew J. Colbrook, George Stepaniants, and Alex Townsend. A complete resolution of Forsythe's conjecture for restarted Conjugate Gradients, 2026. arXiv:2609.04659v1, 4 September 2026.
- [2] Zijian Zeng. Generic characteristic-zero equivalence between derivative Bezout inversion and multipoint evaluation. *Preprints*, August 2026. doi: <https://doi.org/10.20944/preprints202608.2160.v1>.